Math 74: Algebraic Topology

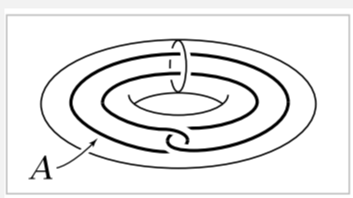
Prishita Dharampal

Problem 1. Show that for a space X , the following three conditions are equivalent:

1. Every map $S^1 \to X$ is homotopic to a constant map, with image a point.
2. Every map $S^1 \to X$ extends to a map $D^2 \to X$.
3. $\pi_1(X, x_0) = 0$ for all $x_0 \in X$.

Deduce that a space X is simply-connected iff all maps $S^1 \to X$ are homotopic. [In this problem, 'homotopic' means 'homotopic without regard to basepoints'.]

Problem 2. Show that there are no retractions $r : X \to A$ where $X = S^1 \times D^2$ and A the circle shown in the figure.



Problem 3. Show that the complement of a finite set of points in $\mathbb{R}^n$ is simply-connected if $n \geq 3$.

Problem 4. Let $X \subset \mathbb{R}^3$ be the union of n lines through the origin. Compute $\pi_1(\mathbb{R}^3 - X)$.

Problem 5. Proof for Theorem 1.38